

instead of preceding directly to Mars, for two main reasons: first, the ships (except for the three detachable torpedo-shaped noses) will lack the streamlining required for flight in the Martian atmosphere; second, it will be more economical to avoid carrying all the fuel needed for the return to earth (which now comprises the bulk of the cargo) all the way down to Mars and then back up again.

Upon reaching the 600-mile orbit—and after some exploratory probings of Mars's atmosphere with unmanned rockets—the first of the three landing craft will be assembled. The torpedo nose will be unhooked, to become the fuselage of a rocket plane. The wings and set of landing skis will be attached, and the plane launched toward the surface of Mars.

The landing of the first plane will be made on the planet's snow-covered polar cap—the only spot where there is any reasonable certainty of finding a smooth surface. Once down, the pioneer landing party will unload its tractors and supplies, inflate its balloonlike living quarters, and start on a 4,000-mile overland journey to the Martian equator, where the expedition's main base will be set up (it is the most livable part of the planet well within the area that scientists want most to investigate). At the equator, the advance party will construct a landing strip for the other two rocket planes. (The first landing craft will be abandoned at the pole.)

In all, the expedition will remain on the planet 15 months. That's a long time—but it still will be too short to learn all that science would like to know about Mars.

When, at last, Mars and the earth begin to swing toward each other in the heavens, and it's time to go back, the two ships that landed on the equator will be stripped of their wings and landing gear, set on their tails and, at the proper moment, rocketed back to the 600-mile orbit on the last leg of the return journey.

What curious information will these first explorers carry back from Mars? Nobody knows—and it's extremely doubtful that anyone now living will ever know. All that can be said with certainty today is this: the trip can be made, and will be made . . . someday.

CHAPTER I

GETTING READY FOR SPACE EXPLORATION

While Wernher von Braun and others during the 1950s were helping to create the public expectation that space travel was just around the corner, the U.S. government was taking the initial tentative steps toward making the United States a spacefaring country. With the development of powerful ballistic missiles for launching nuclear weapons getting under way, it was just a matter of time before variants of those missiles were converted into rockets for launching objects, and, soon after, people into orbit.

The United States would not be first to space, however. American prestige took a serious blow when the Soviet Union launched the first artificial Earth satellite, *Sputnik 1*, on October 4, 1957. The event was seen by many, particularly in the media and the Congress, as proof of Soviet superiority in science, technology, engineering, and social organization—with all the implications that had for the growth of Soviet power and prestige. But *Sputnik 1* did not cause high levels of alarm within the Eisenhower administration. Dwight D. Eisenhower and his associates had a different space priority. To Eisenhower and many of his associates, who had felt the impact of Pearl Harbor, minimizing the risk of a surprise Soviet attack was an overriding concern. The need to see into the Soviet Union and to learn the location and number of its military bases, nuclear facilities, missile facilities, bomber aircraft, and so forth created the demand for strategic reconnaissance on a continental scale. At that time, aircraft overflight and sending camera-carrying balloons over Soviet territory (the latter of which provided little useful intelligence) were the only means available for strategic reconnaissance, and they were forbidden by international law.

The Eisenhower administration grappled with this problem and took a number of different approaches to solve it, including a high-altitude spy plane, the U-2, thought impervious to anti-aircraft attack. Another approach was approving the early development of satellites that could overfly the Soviet Union and return useful information. In 1954 the Air Force began preliminary development of a reconnaissance satellite program, dubbed WS (Weapon System) 117L. This was the first government-approved space program. Although its feasibility was far from certain at the time, it was understood that a satellite could potentially overcome the risks faced by aerial reconnaissance. First, there was no risk of a satellite getting shot down or destroyed in orbit by conventional air defenses. Second, it was not clear under international law if one country's satellite overflying another country's territory in outer space was legal or not. There was no precedent. Establishing the principle of "freedom of space," that is, the acceptability and legality of satellite overflight of another's territory, became a crucial national security issue for the United States.

Meanwhile, as these national defense concerns were being discussed inside the government, the U.S. scientific community independently was developing a proposal for a scientific satellite to be launched during the International Geophysical Year (IGY), which was to run from June 1957 to December 1958. The IGY was a major international effort to study the entire Earth, including its lands, seas, atmosphere, and outer space environments. The international scientific community would collaborate on research and share the results. Sixty-seven countries were to participate, including both the United States and the Soviet Union. To U.S. scientists, orbiting a scientific satellite was a natural extension of their post-World War II research in the upper atmosphere using high-altitude balloons and sounding rockets.

The National Science Foundation approved the National Academy of Sciences' proposal for a scientific satellite. But because of its implications for national security, the satellite program also required approval at the highest levels of the U.S. government. Thus a National Security Council paper, NSC 5520, "Draft Statement of Policy on U.S. Scientific Satellite Program," May 20, 1955, outlined a variety of reasons for approval of the scientific satellite program, including its scientific and technological

benefits, its importance for national prestige, and, most important, its use in establishing the international legal precedent of freedom of space. The satellite's peaceful purposes would be emphasized, since it was to be an unclassified program and the scientific data it acquired would be shared internationally under the sponsorship of the IGY. These factors would increase the chances that no country would protest the overflight of such a satellite. With the legal precedent set, an opening for reconnaissance satellites would be created. Valid scientific interests would provide a convenient screen for equally valid national security concerns. Science would shape the environment for the military to peer behind the Iron Curtain and strengthen U.S. national security. The intentional intertwining of U.S. scientific and national security interests in the U.S. civilian space program was thus part of the U.S. space effort from its inception.

National Security Council, NSC 5520, "Draft Statement of Policy on U.S. Scientific Satellite Program," May 20, 1955

GENERAL CONSIDERATIONS

1. The U.S. is believed to have the technical capability to establish successfully a small scientific satellite of the earth in the fairly near future. Recent studies by the Department of Defense have indicated that a small scientific satellite weighing 5 to 10 pounds can be launched into an orbit about the earth using adaptations of existing rocket components. If a decision to embark on such a program is made promptly, the U.S. will probably be able to establish and track such a satellite within the period 1957-58.
2. The report of the Technological Capabilities Panel of the President's Science Advisory Committee recommended that intelligence applications warrant an immediate program leading to a very small satellite in orbit around the earth, and that re-examination should be made of the principles or practices of international law with regard to "Freedom of Space" from the standpoint of recent advances in weapon technology.
3. On April 16, 1955, the Soviet Government announced that a permanent high-level, interdepartmental commission for interplanetary communications has been created in the

- Astronomic Council of the USSR Academy of Sciences. A group of Russia's top scientists is now believed to be working on a satellite program. In September 1954 the Soviet Academy of Sciences announced the establishment of the Tsiolkovsky Gold Medal which would be awarded every three years for outstanding work in the field of interplanetary communications.
4. Some substantial benefits may be derived from establishing small scientific satellites. By careful observation and the analysis of actual orbital decay patterns, much information will be gained about air drag at extreme altitudes and about the fine details of the shape of and the gravitational field of the earth. Such satellites promise to provide direct and continuous determination of the total ion content of the ionosphere. These significant findings will find ready application in defense communication and missile research. When large instrumented satellites are established, a number of other kinds of scientific data may be acquired . . .
 5. From a military standpoint, the Joint Chiefs have stated their belief that intelligence applications strongly warrant the construction of a large surveillance satellite. While a small scientific satellite cannot carry surveillance equipment and therefore will not have any direct intelligence potential, it does represent a technological step toward the achievement of a large surveillance satellite, and will be helpful to this end so long as the small scientific satellite program does not impede the development of the large surveillance satellite.
 6. Considerable prestige and psychological benefits will accrue to the nation which first is successful in launching a satellite. The inference of such a demonstration of advanced technology and its unmistakable relationship to intercontinental ballistic missile technology might have important repercussions on the political determination of free world countries to resist Communist threats, especially if the USSR were to be the first to establish a satellite. Furthermore, a small scientific satellite will provide a test of the principle of "Freedom of Space." The implications of this principle are being studied within the Executive Branch. However, preliminary studies indicate that

- there is no obstacle under international law to the launching of such a satellite.
7. It should be emphasized that a satellite would constitute no active military offensive threat to any country over which it might pass. Although a large satellite might conceivably serve to launch a guided missile at a ground target, it will always be a poor choice for the purpose. A bomb could not be dropped from a satellite on a target below, because anything dropped from a satellite would simply continue alongside in the orbit.
 8. The U.S. is actively collaborating in many scientific programs for the International Geophysical Year (IGY), July 1957 through December 1958. The U.S. National Committee of the IGY has requested U.S. Government support for the establishment of a scientific satellite during the Geophysical Year. The IGY affords an excellent opportunity to mesh a scientific satellite program with the cooperative world-wide geophysical observational program. The U.S. can simultaneously exploit its probable technological capability for launching a small scientific satellite to multiply and enhance the over-all benefits of the International Geophysical Year, to gain scientific prestige, and to benefit research and development in the fields of military weapons systems and intelligence. The U.S. should emphasize the peaceful purposes of the launching of such a satellite, although care must be taken as the project advances not to prejudice U.S. freedom of action (1) to proceed outside the IGY should difficulties arise in the IGY procedure, or (2) to continue with its military satellite programs directed toward the launch of a large surveillance satellite when feasible and desirable.
 9. The Department of Defense believes that, if preliminary design studies and initial critical component development are initiated promptly, sufficient assurance of success in establishing a small scientific satellite during the IGY will be obtained before the end of this calendar year to warrant a response, perhaps qualified, to an IGY request. The satellite itself and much information as to its orbit would be public information. The means of launching would be classified.

10. A program for a small scientific satellite could be developed from existing missile programs already underway within the Department of Defense. Funds of the order of \$20 million are estimated to be required to give reasonable assurance that a small scientific satellite can be established during 1957-8 . . .

COURSES OF ACTION

11. Initiate a program in the Department of Defense to develop the capability of launching a small scientific satellite by 1958, with the understanding that this program will not prejudice continued research toward large instrumented satellites for additional research or intelligence capabilities, or materially delay other major Defense programs.
12. Endeavor to launch a small scientific satellite under international auspices, such as the International Geophysical Year, in order to emphasize its peaceful purposes, provided such international auspices are arranged in a manner which:
- Preserves U.S. freedom of action in the field of satellites and related programs.
 - Does not delay or otherwise impede the U.S. satellite program and related research and development programs.
 - Protects the security of U.S. classified information regarding such matters as the means of launching a scientific satellite.
 - Does not involve actions which imply a requirement for prior consent by any nation over which the satellite might pass in its orbit, and thereby does not jeopardize the concept of "Freedom of Space."

This statement was approved at a May 26, 1955, meeting of the National Security Council, thereby becoming in effect the first U.S. National Space Policy.

A proposal from the civilian-like Naval Research Laboratory (NRL) for a small scientific satellite developed by the scientific community, funded by the National Science Foundation, and launched on top of a modified sounding rocket, Viking, was selected on September 9, 1955, as the U.S. space contribution to

the IGY. The satellite project's name was Vanguard, and the first launch was expected to be in late 1957 or early 1958.

Vanguard was selected in the face of a competing proposal. The potential existed for the U.S. Army, using a rocket developed by Wernher von Braun and his German rocket team, to launch a satellite in late 1956 or early 1957. The Army Ballistic Missile Agency's (ABMA) proposal was called Project Orbiter. But the Eisenhower administration judged that a U.S. military satellite, launched on a military ballistic missile, before the beginning of the IGY, and without scientific justification, was likely to be viewed as a provocative act by the Soviets, generate a Soviet protest, and thus fail to establish the freedom of space principle. There was also some consideration that it should not be von Braun and his team, so recently employed by Nazi Germany, who should be responsible for this U.S. accomplishment. President Eisenhower decided that the establishment of the freedom of space principle was more important than being first. The Army protested the decision to choose the Navy's Project Vanguard, but to no avail.

Such considerations did not constrain Soviet behavior. Sergei Korolev, the "chief designer" of the Soviet space program, and his associates were determined to be the first into space. On October 4, 1957, the Soviet Union launched the first Earth-orbiting satellite. This "simple satellite," as it was called by Korolev, was explicitly built to beat the United States to the first satellite launch. The Soviets called the satellite "Sputnik," or "fellow traveler," and reported the achievement in a tersely worded press release issued by the official news agency, Tass, printed in the October 5, 1957, issue of the Communist Party newspaper Pravda. The press release ended by linking the satellite launch to the success of the "new socialist society."

"Announcement of the First Satellite," Pravda, October 5, 1957

For several years scientific research and experimental design work have been conducted in the Soviet Union on the creation of artificial satellites of the earth. As already reported in the press, the first launching of the satellites in the USSR were planned for realization in accordance with the scientific research program of

the International Geophysical Year. As a result of very intensive work by scientific research institutes and design bureaus the first artificial satellite in the world has been created. On October 4, 1957, this first satellite was successfully launched in the USSR. According to preliminary data, the carrier rocket has imparted to the satellite the required orbital velocity of about 8000 meters per second. At the present time the satellite is describing elliptical trajectories around the earth, and its flight can be observed in the rays of the rising and setting sun with the aid of very simple optical instruments (binoculars, telescopes, etc.). According to calculations which now are being supplemented by direct observations, the satellite will travel at altitudes up to 900 kilometers above the surface of the earth; the time for a complete revolution of the satellite will be one hour and thirty-five minutes; the angle of inclination of its orbit to the equatorial plane is 65 degrees. On October 5 the satellite will pass over the Moscow area twice—at 1:46 a.m. and at 6:42 a.m. Moscow time. Reports about the subsequent movement of the first artificial satellite launched in the USSR on October 4 will be issued regularly by broadcasting stations. The satellite has a spherical shape 58 centimeters in diameter and weighs 83.6 kilograms. It is equipped with two radio transmitters continuously emitting signals at frequencies of 20.005 and 40.002 megacycles per second (wave lengths of about 15 and 7.5 meters, respectively). The power of the transmitters ensures reliable reception of the signals by a broad range of radio amateurs. The signals have the form of telegraph pulses of about 0.3 second's duration with a pause of the same duration. The signal of one frequency is sent during the pause in the signal of the other frequency. Scientific stations located at various points in the Soviet Union are tracking the satellite and determining the elements of its trajectory. Since the density of the rarefied upper layers of the atmosphere is not accurately known, there are no data at present for the precise determination of the satellite's lifetime and of the point of its entry into the dense layers of the atmosphere. Calculations have shown that owing to the tremendous velocity of the satellite, at the end of its existence it will burn up on reaching the dense layers of the atmosphere at an altitude of several tens of kilometers. As early as the end of the nineteenth century the possibility of realizing cosmic flights by means of rockets was first scientifically substantiated in Russia by the

works of the outstanding Russian scientist K. E. Tsiolkovskii. The successful launching of the first man-made earth satellite makes a most important contribution to the treasure-house of world science and culture. The scientific experiment accomplished at such a great height is of tremendous importance for learning the properties of cosmic space and for studying the earth as a planet of our solar system. During the International Geophysical Year the Soviet Union proposes launching several more artificial earth satellites. These subsequent satellites will be larger and heavier and they will be used to carry out programs of scientific research. Artificial earth satellites will pave the way to interplanetary travel and, apparently, our contemporaries will witness how the freed and conscientious labor of the people of the new socialist society makes the most daring dreams of mankind a reality.

The legality of overflight was suddenly no longer an issue; the United States did not protest Sputnik's overflight of American territory. The international legal precedent had been set. The way was now clear for the United States to develop its reconnaissance satellite program and thereby gain the means to observe the interior of the Soviet Union.

The public and Congress were shocked by the launch of Sputnik. America's sense of security was shattered as Sputnik visibly demonstrated that the United States could be directly threatened by a nation half a world away. The Eisenhower administration had been aware of the imminent launch of the first Soviet satellite and had given some thought to potential public reaction to such an event. But when the launch occurred on October 4, 1957, the administration was surprised by the amount of public concern about the security implications of the Soviet achievement. A crisis atmosphere erupted in the media and in Congress, and although Eisenhower did not agree that the launch of Sputnik warranted an urgent response, he was forced politically into reacting with a series of rapid steps.

Four days after the event, Secretary of State John Foster Dulles sent White House press secretary James Hagerty his suggestions for the text of a press release that would place the Sputnik launch in a nonthreatening context and reassure the public. Although the core of the German "rocket team," including Wernher von Braun, had surrendered to the United States at the end of World

War II, Dulles attributes the Soviet success to German engineers working in the Soviet Union. Although Dulles's comments did not result in a press release, they did form the basis for much of the administration's "official" reaction to the Soviet achievement as well as the core of Eisenhower's comments at a press conference on October 9, after the president had met with his advisers to discuss the significance of Sputnik.

**John Foster Dulles, "Draft Statements on the Soviet Satellite,"
October 8, 1957**

The launching by the Soviet Union of the first earth satellite is an event of considerable technical and scientific importance. However, that importance should not be exaggerated. What has happened involves no basic discovery and the value of a satellite to mankind will for a long time be highly problematical. That the Soviet Union was first in this project is due to the high priority which the Soviet Union gives to scientific training and to the fact that since 1945 the Soviet Union has particularly emphasized developments in the fields of missiles and of outer space. The Germans had made a major advance in this field and the results of their effort were largely taken over by the Russians when they took over the German assets, human and material, at Peenemünde, the principal German base for research and experiment in the use of outer space. This encouraged the Soviets to concentrate upon developments in this field with a use of resources and effort not possible in time of peace to societies where the people are free to engage in pursuits of their own choosing and where public monies are limited by representatives of the people. Despotism which can command the activities and resources of all their people can often produce spectacular accomplishments. These, however, do not prove that freedom is not the best way. While the United States has not given the same priority to outer space developments as has the Soviet Union, it has not neglected this field. It already has a capability to utilize outer space for missiles and it is expected to launch an earth satellite during the present geophysical year in accordance with a program which has been under orderly development over the past two years. The United States welcomes the peaceful achievement of the Soviet

scientists. It hopes that the acclaim which has resulted from their effort will encourage the Soviet Union to seek development along peaceful lines and seek to enrich the spiritual and material welfare of their people. What is happening with reference to outer space makes more than ever important the proposal made by the United States and the other free world members of the Disarmament Subcommittee. I recall my White House statement of August 28 which emphasized the proposal of the Western Powers at London to establish a study group to the end that "outer space shall be used only for peaceful, not military, purposes."

General Andrew Goodpaster, Eisenhower's military aide, took notes during the first discussion between President Eisenhower and his top national security advisers after the Soviet launch of Sputnik 1. Deputy Secretary of Defense Donald Quarles, who among the group was the most well versed on space matters, led the discussion. Eisenhower indicated that he had never worried about being first to space, but he also recognized that the United States had to respond to the Soviet challenge.

**Andrew Goodpaster, "Memorandum of Conference with
the President," October 8, 1957**

Secretary Quarles began by reviewing a memorandum prepared in Defense for the President on the subject of the earth satellite (dated October 7, 1957). He left a copy with the President. He reported that the Soviet launching on October 4th had apparently been highly successful.

The President asked Secretary Quarles about the report that had come to his attention to the effect that Redstone could have been used and could have placed a satellite in orbit many months ago. Secretary Quarles said there was no doubt that the Redstone, had it been used, could have orbited a satellite a year or more ago. The Science Advisory Committee had felt, however, that it was better to have the earth satellite proceed separately from military development. One reason was to stress the peaceful character of the effort, and a second was to avoid the inclusion of materiel, to which foreign scientists might be given access, which is used in our own military rockets. He said that the Army feels

it could erect a satellite four months from now if given the order—this would still be one month prior to the estimated date for the Vanguard. The President said that when this information reaches the Congress, they are bound to ask why this action was not taken. He recalled, however, that timing was never given too much importance in our own program, which was tied to the IGY and confirmed that in order for all scientists to be able to look at the instrument, it had to be kept away from military secrets. Secretary Quarles pointed out that the Army plan would require some modification of the instrumentation in the missile.

He went on to add that the Russians may have in fact done us a good turn, unintentionally, in establishing the concept of freedom of international space—this seems to be generally accepted as orbital space, in which the missile is making an inoffensive passage.

The President asked what kind of information could be conveyed by the signals reaching us from the Russian satellite. Secretary Quarles said the Soviets say that it is simply a pulse to permit location of the missile through radar direction finders . . .

The President asked the group to look ahead five years, and asked about a reconnaissance vehicle. Secretary Quarles said the Air Force has a research program in this area and gave a general description of the project.

Governor Adams recalled that Dr. Pusey had said that we had never thought of this as a crash program, as the Russians apparently did. We were working simply to develop and transmit scientific knowledge. The President thought that to make a sudden shift in our approach now would be to belie the attitude we have had all along. Secretary Quarles said that such a shift would create service tensions in the Pentagon. Mr. Holaday said he planned to study with the Army the back up of the Navy program with the Redstone, adapting it to the instrumentation.

There was some discussion concerning the Soviet request, as to whether we would like to put instruments of ours aboard one of their satellites . . . Several present pointed out that our instruments contain parts which, if made available to the Soviets, would give them substantial technological information.

As a first step in responding to the Soviets, the United States needed to get a satellite into orbit. The first Vanguard launch

attempt on December 6, 1957, ended in a humiliating failure in front of the TV cameras. The Army, having never given up on its ambition to launch a satellite into orbit, had after Sputnik been authorized to attempt a satellite launch, and on January 31, 1958, successfully used von Braun's Jupiter-C rocket (a modified Redstone intermediate range ballistic missile) to put Explorer 1 into orbit.

By February 1958, four months after Sputnik 1, how best to organize for the U.S. response to Sputnik was still uncertain. It was likely that a new space agency would be created; however, its responsibilities, form, and organizational location were still undecided. The question of the military or civilian character of a new agency was discussed in a meeting among the president, vice president, other White House officials, and Republican leaders in Congress. Eisenhower's newly appointed science adviser, James Killian, and Vice President Richard Nixon were advocates of a separate civilian space agency, while the president was undecided on which organizational path to follow. He was also skeptical of carrying out U.S. space projects on a crash basis to compete with Soviet achievements.

L. A. Minnich Jr., "Legislative Leadership Meeting, Supplementary Notes," February 4, 1958

A question was raised as to whether a new Space Agency should be set up within Department of Defense (as provided in the pending Defense appropriation bill), or be set up as an independent agency. The President's feeling was essentially a desire to avoid duplication, and priority for the present would seem to rest with Defense because of paramountcy of defense aspects. However, the President thought that in regard to non-military aspects, Defense could be the operational agent, taking orders from some non-military scientific group. The National Science Foundation, for instance, should not be restricted in any way in its peaceful research.

Dr. Killian had some reservations as to the relative interest and activity of military vs. peaceful aspects, as did the Vice President who thought our posture before the world would be better if non-military research in outer space were carried forward by an agency entirely separate from the military.

There was some discussion of the prospect of a lunar probe. Dr. Killian thought this might be next on the list of Russian efforts. He had some doubt as to whether the United States should at this late date attempt to press a lunar probe, but the question would be fully canvassed by the Science Advisory Committee in the broad survey it had under way. Dr. Killian thought the United States might do a lunar probe in 1960, or perhaps get to it on a crash program by 1959. Sen. Saltonstall had heard, however, that it might even be accomplished in 1958, if pressed hard enough.

The President was firmly of the opinion that a rule of reason had to be applied to these Space projects—that we couldn't pour unlimited funds into these costly projects where there was nothing of early value to the Nation's security. He recalled the great effort he had made for the Atomic Peace Ship but Congress would not authorize it, even though in his opinion it would have been a very worthwhile project.

And in the present situation, the President mused, he would rather have a good Redstone [a missile carrying a nuclear warhead] than be able to hit the moon, for we didn't have any enemies on the moon.

Sen. Knowland pressed the question of hurrying along with a lunar probe, because of the psychological factor. He recalled the great impact of Sputnik, which seemed to negate the impact of our large mutual security program. If we are close enough to doing a probe, he said, we should press it. The President thought it might be OK to go ahead with it if it could be accomplished with some missile already developed or nearly ready, but he didn't want to just rush into an all-out effort on each one of these possible glamor performances without a full appreciation of their great cost. Also, there would have to be a clear determination of what agency would have the responsibility.

The Vice President reverted to the idea of setting up a separate agency for "peaceful" research projects, for the military would be deterred from things that had no military value in sight. The President thought Defense would inevitably be involved since it presently had all the hardware, and he did not want further duplication. He did not preclude having eventually a great Department of Space.

The Eisenhower administration also began organizing the U.S. government for the space age. A first action, taken soon after the Soviet Union on November 3, 1957, launched the large Sputnik 2 carrying the dog Laika, was to create both a White House position of presidential science adviser and also a part-time advisory group of eminent scientists called the President's Science Advisory Committee (PSAC). One of PSAC's initial assignments was to examine the rationales for space activity and recommend how to organize the United States for space. After much deliberation and consideration of various alternatives, three separate space activity paths were developed. One track included military space activities. On February 7, 1958, these activities were assigned to a newly created Advanced Research Projects Agency (ARPA) within the Department of Defense. Eisenhower hoped that establishing this new agency with authority over all military space efforts would prevent intense interservice rivalry over the space mission. Another track approved the same day was a top secret strategic reconnaissance program to observe the Soviet Union, called CORONA; this program was based on one element of WS-117L, a photo intelligence satellite that would drop exposed film from orbit, where it would be recovered in midair. Management of CORONA was assigned to a covert CIA and Air Force team similar to that managing the U-2 spy aircraft program; this team subsequently would become the core of the secret National Reconnaissance Office.

The third track, chosen after a careful examination of alternatives, was to create a separate civilian space program, focused on science and exploration activities, conducted on an unclassified basis, and open to international cooperation, with a new civilian space agency to manage it. It is worth noting that based upon the debates and decisions made in the immediate aftermath of Sputnik, the U.S. government space program was divided into the civilian, military, and intelligence space sectors that still exist today, more than sixty years later.

There are many reasons why the United States decided to create a separate civilian space agency. Initially, President Eisenhower reasoned that the primary focus of the space program would be military and intelligence missions, and that a separate civilian agency might not be necessary. But he was soon

convinced that a civilian space program was important both for the scientific aspects of space and for domestic and international audiences. The United States could demonstrate how open, scientific, and peaceful its purposes in space were, as opposed to the closed, militaristic nature of the Soviet program. Furthermore, the U.S. scientific community did not trust the U.S. military to have its best interests in mind when it made its budgetary and priority decisions. Eisenhower agreed. Besides, the potential for international cooperation in space science would be unduly constrained in a scientific program under military control. Finally, putting a civilian face on American space activities would help divert attention from Eisenhower's primary goal for the space program: getting strategic intelligence on the Soviet Union.

With the requirement for a civilian space agency decided, the next question was how to organize it. With the military option ruled out, three primary alternatives were considered. The first option was to create a new government agency entirely from scratch. But it would take too much time to legislate, fund, organize, and build new facilities. The second possibility was to assign civilian space activities to an expanded Atomic Energy Commission (AEC). But the AEC had virtually no experience with space-related programs, and civilian space activities would interfere with its current, vital national security function. It was finally agreed that expansion of the National Advisory Committee for Aeronautics (NACA) offered the best solution. Although it did not have extensive rocket or space experience and capabilities, NACA had a number of factors in its favor. NACA was not part of the Department of Defense. It had been working on flight technology since its inception during World War I, it had a history of successfully managing interactions with the military and industry, it had an extensive network of research installations across the United States, and it had developed a preliminary proposal for a comprehensive, well-reasoned space program. By early March 1958, draft legislation was in the works for the creation of a space agency built upon the NACA structure.

The following memorandum was the culmination of discussion on how best to organize the nation's space effort, and it laid the basis for President Eisenhower's decision to create a new civilian space agency based on NACA.

James R. Killian Jr. (Special Assistant for Science and Technology), Percival Brundage (Director, Bureau of the Budget), Nelson A. Rockefeller (Chairman, President's Advisory Committee on Government Organization), Memorandum for the President, "Organization for Civil Space Programs," March 5, 1958, with attached: "Summary of Advantages and Disadvantages of Alternative Organizational Arrangements"

THE PROBLEM

As you know, there will soon be presented for your consideration civil space programs for the United States which will entail increased expenditures and the employment of important numbers of scientists, engineers and technicians.

This Committee, in conjunction with the Director of the Bureau of the Budget and your Special Assistant for Science and Technology, have given consideration to the manner in which the executive branch should be organized to conduct the new program. This memorandum contains our joint findings and recommendations. The memorandum (1) discusses some of the factors which should be taken into account in establishing the government's organization for these civil space programs, (2) recommends a pattern of organization, and (3) indicates certain interim actions which will be necessary. Also attached is a summary of the advantages and disadvantages of certain alternative organizational arrangements.

Discussions to date suggest that an aggressive space program will produce important civilian gains in the form of advances in general scientific knowledge and protection of the international prestige of the United States. These benefits will be in addition to such military uses of outer space as may prove feasible.

ESTABLISHING A LONG TERM ORGANIZATION

Because of the importance of the civil interest in space exploration, the long term organization for Federal programs in this area should be under civilian control. Such civilian domination is also suggested by public and foreign relations considerations. However, civilian control does not envisage taking out from military control projects relating to missiles, anti-missile defense, reconnaissance